Appendix A. Explicit compliance and stiffness kernels in the Hankel domain

This appendix collects the explicit transform-domain kernels used in the main text. We consider two interfacial idealizations for the TI layer of thickness d: (i) a perfectly bonded interface, for which the surface displacements are related to the applied tractions by the compliance functions $\mathcal{F}_{ij}(\xi)$ [cf. Eqs. (19)–(20)] and, equivalently, by the stiffness kernels $\mathcal{K}_{ij}(\xi)$ [cf. Eqs. (36)–(37)]; and (ii) a frictionlessly bonded interface, for which the corresponding kernels are denoted by $\bar{\mathcal{F}}_{ij}(\xi)$ and $\bar{\mathcal{K}}_{ij}(\xi)$ [cf. Eq. (51)]. All kernels satisfy the symmetry relations $\mathcal{F}_{ij} = \mathcal{F}_{ji}$ and $\mathcal{K}_{ij} = \mathcal{K}_{ji}$ (and analogously for the barred quantities). For compactness, we report only the independent components in each case.

Appendix A.1. Perfectly bonded interface: $\mathcal{F}_{ij}$ and $\mathcal{K}_{ij}$

Specifically, the compliance functions $\mathcal{F}_{ij}(\xi)$ are given by Eq. (A.1), with the auxiliary coefficients α_1–α_5 defined immediately below that equation. The associated stiffness kernels $\mathcal{K}_{ij}(\xi)$ are also shown in Eq. (A.2), with β_1 and β_2 defined in Eq.(A.3).

$$
\begin{aligned}
\mathcal{F}_{zz} &= \frac{C_{11}\left(s_1^2-s_2^2\right)\left(s_1\left(C_{44}-C_{33}s_2^2\right)\sinh\left[\left(s_1-s_2\right)d\xi\right]+\left(s_1-s_2\right)\left(C_{44}+C_{33}s_1s_2\right)\sinh\left(s_2d\xi\right)\cosh\left(s_1d\xi\right)\right)}{\xi\left(C_{13}^2-C_{11}C_{33}\right)\left(\alpha_1-\alpha_2\left(s_1-s_2\right)^2\sinh\left(s_1d\xi\right)\sinh\left(s_2d\xi\right)-\alpha_3\cosh\left[\left(s_1-s_2\right)d\xi\right]\right)},\\
\mathcal{F}_{zr} &= \frac{\left[2s_1^2s_2^2\alpha_4-\left(s_1^2+s_2^2\right)\alpha_5\right]\left(\cosh\left[\left(s_1-s_2\right)d\xi\right]-1\right)-\left(s_1-s_2\right)^2\left(s_1s_2\alpha_4+\alpha_5\right)\sinh\left(s_1d\xi\right)\sinh\left(s_2d\xi\right)}{\xi\left(C_{13}^2-C_{11}C_{33}\right)\left(\alpha_1-\alpha_2\left(s_1-s_2\right)^2\sinh\left(s_1d\xi\right)\sinh\left(s_2d\xi\right)-\alpha_3\cosh\left[\left(s_1-s_2\right)d\xi\right]\right)},\\
\mathcal{F}_{rr} &= \frac{C_{33}\left(s_1^2-s_2^2\right)\left(\sinh\left[\left(s_1-s_2\right)d\xi\right]s_1\left(C_{11}-C_{44}s_2^2\right)+\sinh\left(s_2d\xi\right)\cosh\left(s_1d\xi\right)\left(s_1-s_2\right)\left(C_{11}+C_{44}s_1s_2\right)\right)}{\xi\left(C_{13}^2-C_{11}C_{33}\right)\left(\alpha_1-\alpha_2\left(s_1-s_2\right)^2\sinh\left(s_1d\xi\right)\sinh\left(s_2d\xi\right)-\alpha_3\cosh\left[\left(s_1-s_2\right)d\xi\right]\right)},
\end{aligned}
\tag{A.1}
$$

where $\alpha_1 = 2s_1^2s_2^2\left(C_{13}+C_{44}\right)$, $\alpha_2 = C_{11}+C_{44}s_1s_2$, $\alpha_3 = C_{11}\left(s_1^2+s_2^2\right)-2C_{44}s_1^2s_2^2$, $\alpha_4 = C_{13}C_{44}-C_{11}C_{33}$, and $\alpha_5 = C_{11}\left(C_{13}-C_{44}\right)$.

$$
\begin{aligned}
\mathcal{K}_{zz} &= \frac{C_{33}C_{44}\xi\left(s_1^2-s_2^2\right)\left(\left(C_{11}-C_{44}s_2^2\right)s_1\sinh\left[\left(s_1+s_2\right)d\xi\right]-\left(C_{11}-C_{44}s_1s_2\right)\left(s_1+s_2\right)\sinh\left(s_2d\xi\right)\cosh\left(s_1d\xi\right)\right)}{\beta_1\left(1-\cosh\left[\left(s_1-s_2\right)d\xi\right]\right)+\beta_2\sinh\left(s_1d\xi\right)\sinh\left(s_2d\xi\right)},\\
\mathcal{K}_{zr} &= \frac{C_{44}\xi\left(\left[2s_1^2s_2^2\alpha_4-\left(s_1^2+s_2^2\right)\alpha_5\right]\left(1-\cosh\left[\left(s_1-s_2\right)d\xi\right]\right)+\left(s_1-s_2\right)^2\left(s_1s_2\alpha_4+\alpha_5\right)\sinh\left(s_1d\xi\right)\sinh\left(s_2d\xi\right)\right)}{\beta_1\left(1-\cosh\left[\left(s_1-s_2\right)d\xi\right]\right)+\beta_2\sinh\left(s_1d\xi\right)\sinh\left(s_2d\xi\right)},\\
\mathcal{K}_{rr} &= \frac{C_{11}C_{44}\left(s_1^2-s_2^2\right)\xi\left(s_2\left(C_{44}s_1^2-C_{11}\right)\sinh\left[\left(s_1-s_2\right)d\xi\right]+\left(s_1-s_2\right)\left(C_{44}s_1s_2+C_{11}\right)\sinh\left(s_2d\xi\right)\cosh\left(s_1d\xi\right)\right)}{\beta_1\left(1-\cosh\left[\left(s_1-s_2\right)d\xi\right]\right)+\beta_2\sinh\left(s_1d\xi\right)\sinh\left(s_2d\xi\right)},
\end{aligned}
\tag{A.2}
$$

where

$$
\begin{aligned}
\beta_1 &= -2\left(s_1^2+s_2^2\right)C_{11}C_{44}+2s_1^2s_2^2\left(C_{11}C_{33}+C_{44}^2\right),\\
\beta_2 &= \left(s_1-s_2\right)^2\left[2C_{11}C_{44}+s_1s_2\left(C_{11}C_{33}+C_{44}^2\right)\right].
\end{aligned}
\tag{A.3}
$$

Appendix A.2. Frictionlessly bonded interface: $\bar{\mathcal{F}}_{ij}$ and $\bar{\mathcal{K}}_{ij}$

For the frictionlessly bonded interface, the corresponding compliance kernels $\bar{\mathcal{F}}_{ij}(\xi)$ are given in Eq. (A.4).

$$
\begin{aligned}
\bar{\mathcal{F}}_{zz} &= \frac{(s_1^2 - s_2^2)\, s_1 s_2\, C_{33}\, \tanh(s_1 d\xi)\, \tanh(s_2 d\xi)}{(C_{11}C_{33} - C_{13}^2)\, \xi \,[\, s_1 \tanh(s_1 d\xi) - s_2 \tanh(s_2 d\xi)\,]}, \\
\bar{\mathcal{F}}_{zr} &= -\frac{s_1(C_{13} + C_{33}s_2^2) \tanh(s_1 d\xi) - s_2(C_{13} + C_{33}s_1^2) \tanh(s_2 d\xi)}{(C_{11}C_{33} - C_{13}^2)\, \xi \,[\, s_1 \tanh(s_1 d\xi) - s_2 \tanh(s_2 d\xi)\,]}, \\
\bar{\mathcal{F}}_{rr} &= \frac{(s_1^2 - s_2^2)\, C_{33}}{(C_{11}C_{33} - C_{13}^2)\, \xi \,[\, s_1 \tanh(s_1 d\xi) - s_2 \tanh(s_2 d\xi)\,]}.
\end{aligned} \tag{A.4}
$$

And the associated stiffness kernels $\bar{\mathcal{K}}_{ij}(\xi)$ are reported in Eq. (A.5).

$$
\begin{aligned}
\bar{\mathcal{K}}_{zz} &= \frac{s_1 s_2 (s_1^2 - s_2^2)\, \xi\, C_{33} C_{44}}{s_1(C_{11} - C_{44}s_2^2)\, \tanh(s_2 d\xi) - s_2(C_{11} - C_{44}s_1^2)\, \tanh(s_1 d\xi)}, \\
\bar{\mathcal{K}}_{zr} &= -\frac{\xi C_{44}\big[s_1(C_{11} + C_{13}s_2^2)\, \tanh(s_2 d\xi) - s_2(C_{11} + C_{13}s_1^2)\, \tanh(s_1 d\xi)\big]}{s_1(C_{11} - C_{44}s_2^2)\, \tanh(s_2 d\xi) - s_2(C_{11} - C_{44}s_1^2)\, \tanh(s_1 d\xi)}, \\
\bar{\mathcal{K}}_{rr} &= \frac{C_{11} C_{44}\, \xi\, (s_1^2 - s_2^2) \tanh(s_1 d\xi) \tanh(s_2 d\xi)}{s_1(C_{11} - C_{44}s_2^2)\, \tanh(s_2 d\xi) - s_2(C_{11} - C_{44}s_1^2)\, \tanh(s_1 d\xi)}.
\end{aligned} \tag{A.5}
$$